2 Docker and the install

FULL

02. Install Docker

You do not need to do this by hand: the install (menu 1) installs Docker for you. This guide only explains it.

Quick version

No time to read it all? This is the short version. The full guide is below.

- This guide installs Docker on your Ubuntu laptop. Docker runs programs in sealed boxes called containers.
- It takes 30 minutes and is free.
- Steps 1 to 3 follow Docker's own page: remove old pieces, add Docker's software source, then install Docker and Compose. If Docker changes their steps, trust their page.
- Step 4: `sudo usermod -aG docker $USER`, then **log out and log back in**. This step is easy to forget.
- Test it: `docker run --rm hello-world`. You should see "Hello from Docker!".
- `permission denied ... docker.sock` means you did not log out and back in after step 4.

What you will build

Docker on your Ubuntu laptop. Docker runs programs in sealed boxes called **containers**. Every project after this one is "download a container, start it". It keeps your laptop tidy, and

you can delete a project without leaving a mess.

Time and cost

30 minutes. Free.

What you need first

- 01 Ubuntu done, and internet working.

Steps

This follows Docker's own instructions for Ubuntu (https://docs.docker.com/engine/install/ubuntu/). If Docker changes their steps, their page is the one to trust.

1. Open a terminal. Remove any old Docker pieces (it is fine if it says none are installed):

```
sudo apt remove -y docker.io docker-compose docker-compose-v2 docker-doc podman-
docker containerd runc
```

2. Add Docker's official software source:

```
sudo apt update
sudo apt install -y ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o
/etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
echo "deb [arch=$(dpkg --print-architecture) signed-
by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu $(.
/etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}") stable" | sudo tee
/etc/apt/sources.list.d/docker.list > /dev/null
sudo apt update
```

3. Install Docker and Compose:

```
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin
docker-compose-plugin
```

4. Let your user run Docker without typing `sudo` every time:

```
sudo usermod -aG docker $USER
```

Then **log out and log back in** (or restart). This step is easy to forget.

5. Test it:

```
docker run --rm hello-world
```

You should see: "Hello from Docker!" and some text.

6. Test Compose:

```
docker compose version
```

You should see: a version number.

Six commands you will use all the time

Command	What it does
`docker ps`	Lists containers that are running now
`docker ps -a`	Lists all containers, including stopped
`docker logs -f NAME`	Shows what a container is saying. `Ctrl+C` to stop watching
`docker compose up -d`	Starts everything in the compose file in the current folder

Command	What it does
`docker compose down`	Stops and removes those containers (your data folders stay)
`docker compose pull`	Downloads newer versions

How you know it worked

`docker run --rm hello-world` printed the hello message, with no `sudo`.

If something goes differently

What you see	What to do
`permission denied ... docker.sock`	You did not log out and back in after step 4. Do that.
`Unable to locate package docker-ce`	Step 2 did not finish. Run its commands again, one at a time.
`E: Could not get lock`	Another update is running. Wait 2 minutes and try again.
The `echo "deb ..."` line fails	Ubuntu is very new and Docker may not have it yet. Ask Gemini: "docker.com has no repo for my Ubuntu version, what do I do?" and paste the error.

Make it yours

- Run `docker run --rm -it ubuntu bash` to step inside a throwaway Ubuntu. Type `exit` to leave. It is gone forever when you leave.
- Run `docker system df` to see how much space Docker uses. To clean up Docker's leftovers safely, use menu `M`, `14`, then `5` (or `bash garden.sh cleanup`): it removes only the garden's own old versions, older than 7 days, and asks first.

Next: 03 Run your garden website